AGENTHON 2026

Track 2 V7-V8 Research & Development Report

From the Limits of the Legacy Stack to Text-Gated Mixture-of-Experts

V7: legacy-final / transition stage
V8: clean-reset architecture, Development ablations, and structural research branches

Report date: 2026-10-07

Internal research record / reproducibility note

Executive Summary

Bottom line
The V7 stage exhausted the incremental PCA2/GARCH/Copula/EVT and weak text-tail paradigm without producing a robust sealed-Development edge. V8 reset the stack and made House text capable of moving the predictive distribution materially. Text-ON recovered to 0.9993, but the text-OFF numerical backbone collapsed to 2.4718. The main conclusion is therefore not merely that text matters, but that text should route market states or experts on top of a stable numerical floor.

- V7 naming in this report: no standalone V7 package is preserved in the archived project files. To avoid fabricating an architecture, V7 denotes the terminal legacy/transition stage consisting of the V6 production candidate, PCA2-only ablation, and the reset decision.
- Core V7 Development evidence: V6 = 1.0366 and PCA2-only = 1.0366. Local pseudo-OOS gains did not survive sealed Development.
- Core V8 structure: 5-day multivariate block bootstrap, 60/40 recent/full-history blend, 1,000 coherent joint draws, and House scenarios controlling probability, shift_sd, and vol_scale.
- V8 Text-ON Development = 0.9993. This recovered the prior 1.0366 loss but did not establish a meaningful advantage over the normalized reference level of 1.0.
- V8 Text-OFF Development = 2.4718. Removing House exposed a very weak numerical continuation engine.
- DNS, F2 transfer/router, F4 numerical shock gate, and F1 analog sampling all failed the precommitted broad-generalization criteria.
- Next direction (V9): use an official/reference-like continuation floor and let House extract repricing/shock state evidence that softly routes probability mass across expert distributions.

Version naming note

The archive formally documents V1-V6 and then the V8 Clean Start. No standalone V7 package or V7-specific report is preserved. To maintain factual continuity, this report uses "V7" as the label for the terminal pre-reset legacy/transition stage: the V6 production candidate, the PCA2-only Development ablation, and the 2026-10-06 reset decision. No separate V7 algorithm is invented.

1. Competition frame and evaluation objective

Track 2 requires a joint predictive distribution represented by hundreds or thousands of Monte Carlo draws rather than a single point forecast. The score combines 50% Marginal CRPS, 30% Joint Variogram, and 20% Tail Penalty, with lower values better. A useful model must therefore get location, dispersion, extremes, and cross-asset co-movement right at the same time.

Component	Weight	Modeling implication
Marginal CRPS	50%	Location and dispersion of each asset-horizon marginal
Joint Variogram	30%	Cross-asset / cross-horizon coherence within the same draw
Tail Penalty	20%	Calibration of extreme quantiles such as 1%, 5%, 95%, and 99%

2. V7 - terminal legacy and transition stage

2.1 Starting point and modeling philosophy

The central V7-stage question was whether a progressively refined numerical engine could create a sealed-Development advantage. Across V1-V6 the project tested EWMA volatility, block bootstrap, PCA yield-curve factors, Student-t/GARCH variants, copula dependence, EVT/POT tails, and a House text-tail overlay.

- Strength: several local walk-forward experiments improved marginal and tail metrics versus the reference.
- Bottleneck: Joint Variogram frequently deteriorated and offset the marginal/tail gains.
- Text usage: in V4/V6, text mainly acted as a weak 5-10% tail-mixture overlay.
- Terminal production candidate: PCA2-based numerical engine with an optional House text-tail path.

2.2 Validation and Development evidence

Experiment/model	Result	Interpretation
Local PCA2 pseudo-OOS	~9% better vs Reference	Promising locally, failed to transfer
Development V6	1.0366	Worse than normalized 1.0 reference
Development PCA2-only	1.0366	Same as V6; weak text-tail added no measurable sealed gain
F4 EVT research	Tail ~10.4% better; Composite ~0.72% better	Component gain too small for overall gap
Robust GARCH / copula	Mostly ~0-1% local changes	Parameter refinement not structural

Decisive V7 failure
V6 and PCA2-only produced exactly the same sealed Development loss of 1.0366. This showed that the text-tail overlay did not move the official distribution materially and that local pseudo-OOS ranking could not be extrapolated to sealed Development.

2.3 Lessons carried out of V7

- When numerical complexity outruns predictive information, parameter instability can dominate performance, as seen in Student-t/GJR variants.
- Incremental PCA/copula/EVT refinements can improve individual components without explaining the leaderboard gap.
- Text should not be a tail decoration; it must be able to change the centre, width, and regime/state transition of the distribution.
- Development attempts should be reserved for hypotheses that materially change the forecast distribution, not 0.x% local tweaks.

3. V8 - clean reset and new predictive mechanisms

3.1 Reset principles

On 2026-10-06, V8 restarted from the official repository rather than inheriting the V1-V6/PCA/GARCH/EVT/Copula forecasting logic. Legacy models were retained only as reproducible research archives. Reused assets were plumbing: I/O, cutoff/horizon semantics, output schema, Docker/packaging, and the House API client.

3.2 First submitted V8 architecture

Block	V8 v0 implementation
Numerical generator	Multivariate 5-day block bootstrap
Regime blend	60% most recent 252 observations + 40% full history
Draws	1,000 coherent joint trajectories
House output	1-4 scenarios + probabilities + per-asset shift_sd + vol_scale
Joint coherence	One draw index represents one market scenario shared across assets/horizons
Family prompts	Specialized F1 continuation / F2 regime shift / F3 cross-asset / F4 tail-shock prompts
Fallback	If House is unavailable: continuation p=1, shift=0, vol=1

$$P(Y) = \Sigma_s \, p_s \cdot P_s(Y \mid \text{numerical history, shift_sd, vol_scale})$$

3.3 Structural and engineering validation

- Text-OFF structural validation across 103 public practice units excluding the example: 103/103 passed.
- g0-g3 gates, finite outputs, row counts, and cutoff semantics passed.
- Docker linux/amd64 build and network-disabled execution succeeded.
- Mock-House ablations verified the reasoning path and distribution-change plumbing.
- This stage established reproducible submission infrastructure, not predictive accuracy.

3.4 Development results: Text-ON versus Text-OFF

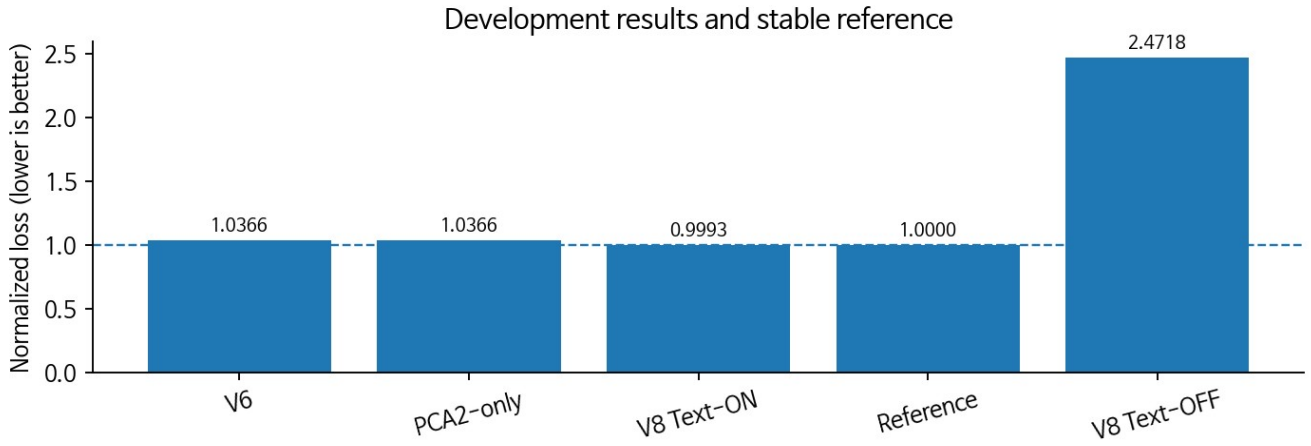


Figure 1. Normalized loss magnitude corresponding to the displayed leaderboard score. Lower is better.

Submission	Displayed score	Loss used here	Interpretation
V6	-1.0366	1.0366	Legacy final candidate; worse than reference
PCA2-only	-1.0366	1.0366	No measurable text-tail effect

V8 Text-ON	-0.9993	0.9993	Recovered to reference level; no clear edge
V8 Text-OFF	-2.4718	2.4718	Numerical continuation backbone collapsed

Key interpretation
The 2.4718 Text-OFF result should not be read as 1.47 units of pure text alpha. The more direct interpretation is that the V8 numerical backbone was very weak and House materially repaired the centre/width/scenario distribution. The next model should therefore not use V8 OFF as its continuation expert.

3.5 V8 structural research branches

Branch	Hypothesis	Key result	Decision
DNS v0	UST Level/Slope/Curvature factor bootstrap	Mostly 0.x% effects; cut-sizing Variogram deteriorated roughly 10x	Reject
F2 static transfer	G10 -> sparse EM target ridge transfer	Some CNY/INR cases improved 5-50%, BRL deteriorated	Partial
F2 state/router	Route among raw/static/state using prior historical folds	Mean 0.8865 but median 1.0000 and holdout win rate 16.7%	Reject
F4 shock gate	Numerical state -> p_shock -> normal/shock mixture	0/15 strong; median Tail 1.0112, CRPS 1.0057	Reject
F1 analog	Condition on historical origins with similar current state	20 units: median ratio 1.0379, unit win 25%, strong 4/20	Reject

3.6 Branch-level interpretation

- DNS: the economically interpretable curve-factor representation remained attractive, but fixed-lambda factor-change bootstrap did not beat raw tenor history robustly.
- F2 transfer: cross-market information can be very strong in specific regimes. However, the spectacular CNY-stress average gain disappeared in temporal holdout, and no generalizable router emerged.
- F4 shock gate: numerical state features alone did not provide a broad shock-probability solution; some funding-stress units suffered large tail deterioration.
- F1 analog: nonparametric state analogs improved a few cards by 10-20% but worsened the typical unit and showed large worst-fold instability.
- Common pattern: rearranging or re-expressing the same numerical history was not enough to create the large information difference needed on sealed Development.

4. V7 versus V8

Dimension	V7	V8	Conclusion
Starting mode	Incremental refinement of the existing numerical stack	Clean reset from official repository	V8 reduced model-selection-overfit risk
Role of text	Weak tail-weight adjustment	Scenario probability, centre, and width changed directly	Text had much larger operational influence in V8
Numerical backbone	PCA2 / GARCH / copula / EVT family	5-day multivariate block bootstrap	V8 OFF exposed backbone weakness
Development	1.0366	Text-ON 0.9993 / OFF 2.4718	ON recovered; OFF collapsed
Research conclusion	Incremental numerical tuning saturated	Numerical rearrangements still failed to generalize	Need stable floor + text state routing

5. From V8 to V9: Text-Gated Probabilistic Mixture-of-Experts

The most important V8 result was dual: House could move the distribution materially, yet the V8 numerical continuation engine was not a viable standalone baseline. V9 therefore moves from "House repairs a weak model" to "a stable baseline is the default odds, and text reallocates probability mass to other expert distributions only when new information justifies repricing."

$$P(Y \mid H,T) = \pi_C(H,T)\, P_C(Y \mid H) + \pi_R(H,T)\, P_R(Y \mid H,D) + \pi_S(H,T)\, P_S(Y \mid H,D)$$

Expert	Role	V9 v0 principle
Continuation	Stable normal-state floor	Official/reference-like joint forecast
Repricing	Central repricing after policy/inflation/growth information	Historical joint repricing events aligned with the text-implied direction
Shock	Rare/extreme state	Text decides shock activation; history supplies shock magnitude and shape

- House does not directly predict exact basis points, returns, or mixture probabilities.
- House structures repricing_evidence, shock_evidence, direction, affected assets, and confidence.
- A deterministic router converts the evidence to mixture weights.
- Each draw selects one global expert state shared across all assets/horizons.
- Validation must compare Continuation-only vs Static MoE vs Text-Gated MoE.

6. Final research lessons and operating rules

1. Local pseudo-OOS is an idea filter, not a proxy leaderboard for sealed Development.
2. A 0.x% local gain is a robustness signal, not a submission thesis.
3. Text ablations are interpretable only when the numerical baseline is independently stable.
4. Text is easier to validate as market-state inference and model routing than as direct numerical forecasting.
5. In a mixture model, do not elementwise-average expert forecasts; preserve multimodality with draw-level categorical mixing.
6. For F3, the same macro state must be shared across all assets/horizons within a draw.

7. Archive failed branches instead of repeatedly rescuing them with parameter tuning.
8. Use the next Development attempt only after a reference-like floor plus text-gated expert routing passes structural validation and ablation.

Appendix. Experiment timeline and key numbers

Date	Event	Key number	Conclusion
2026-10-06	V6 Development	1.0366	Legacy production candidate
2026-10-06	PCA2-only Development	1.0366	Text-tail no sealed effect
2026-10-06	V8 clean reset	-	Legacy numerical family retired
2026-10-07	V8 Text-ON Development	0.9993	Recovered to reference level
2026-10-07	F2 router holdout	median 1.0000	Win rate 16.7%; reject
2026-10-07	F4 shock gate	median tail 1.0112	0/15 strong; reject
2026-10-07	F1 analog	median 1.0379	Unit win 25%; reject
2026-10-07	V8 Text-OFF Development	2.4718	Numerical backbone rejected
2026-10-07	V9 design frozen	-	Reference continuation + text-gated Repricing/Shock

Internal source record

- Agenthon_T2_Retrospective_Reset_2026-10-06.pdf
- 00_Agenthon_T2_V8_Clean_Start_Handoff.pdf
- Agenthon_T2_F1-F4_Model_Ideas_and_Papers.pdf
- V8 DNS / F2 Transfer / F4 Shock Gate / F1 Analog experiment outputs, 2026-10-07
- CodaBench Development submissions: V6, PCA2-only, V8 Text-ON, V8 Text-OFF
- Official Track 2 public repository and qfbench2-common 2.4.4 rules

Reproducibility note
"V7" in this report is a label for the terminal pre-reset legacy/transition stage, not a separately preserved package. This convention is explicit so that no unarchived V7 architecture is invented.